



INTRODUCTION TO BIG DATA ANALYSIS IEE3593.02-00



PREDICTING RETAIL SPENDING IN SOUTH KOREA

Presented by Team Data Blazers
(Team 9)

MEET THE BLAZERS



LIU ZIHAO
(JEFF)



TSE MAN KI
(BRIAN)



SHASHANK
VEMPARALA
(PRABHAT)



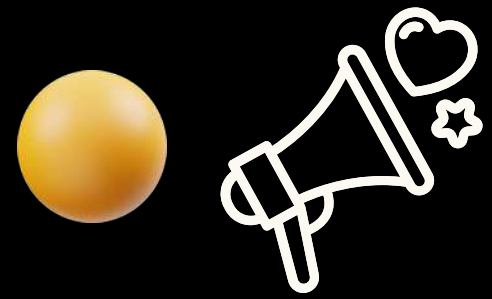
MUHAMMAD
RAFII
MUNFAADI



SIQI ZHONG
(BELLA)



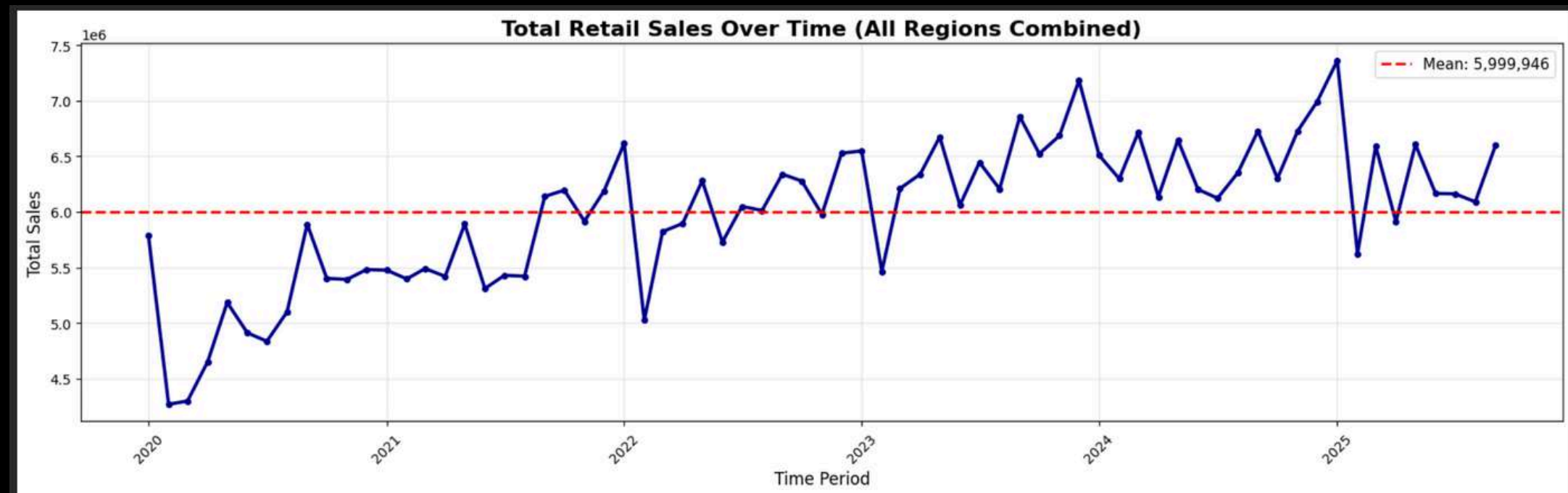
ASK A QUESTION (TOPIC & MOTIVATION)



Unpacking Regional Retail Performances in South Korea

Why did we select this?

South Korea has an aging population, and according to another team covering fertility rates, fertility rates have also been declining. However, according to our EDA, retail sales are increasing over time. Why and how? What is driving this increase?



ASK A QUESTION (TOPIC & MOTIVATION)

Motivation (Practical Significance):

- For Businesses & Retailers: Informs data-driven decisions on location planning, inventory management, and regional marketing strategies.
 - For Policymakers: Helps design targeted regional policies, such as employment support or economic stimulus, where they are needed most.
 - For Socio-Economic Planning: Provides insights into how long-term trends like aging populations and youth unemployment may reshape regional economic landscapes.
- 

GET DATA



KOSIS

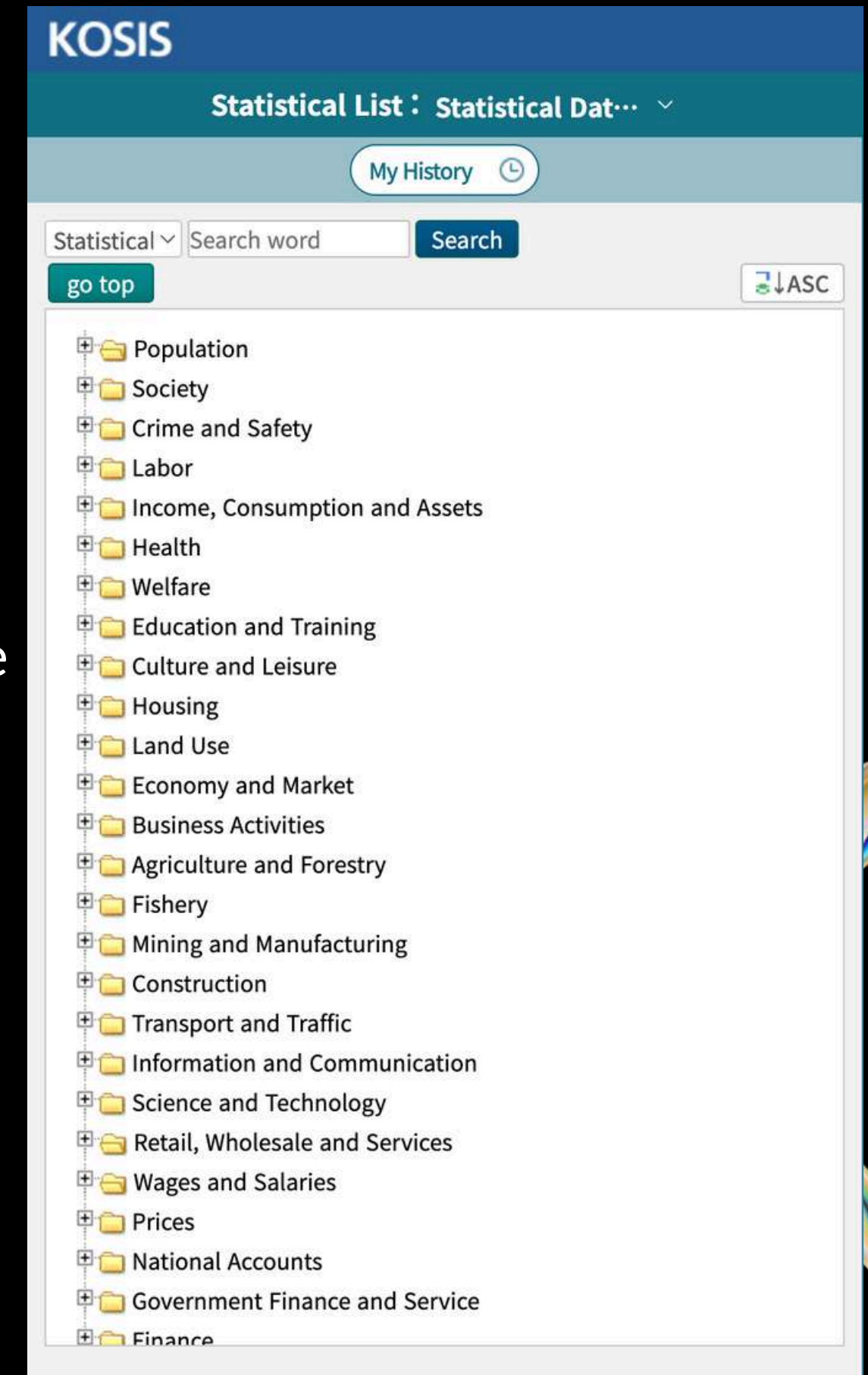
LINK: KOSIS.KR

Why KOSIS?

We tried to work on province-wise monthly data, and KOSIS was one of the only sources that had data broken down by provinces for a big time period

ABOUT OUR DATASETS

- All our data files were downloaded
- All the datasets are in CSV format
- Temporal Resolution - Monthly / Quarterly converted to monthly in the pre-processing
- Spatial Resolution - Regional/Provincial level (Covers 17 regions across South Korea)



OUR PROBLEM AFTER INITIAL PRESENTATION

1. Data is too small, add more features to explain retail dynamics in temporal manner.
3. You will need to collect more diverse data. Right now, the data seems to be insufficient



Retail Data

```
--- Data Volume ---  
Shape of retail_df: (72, 18) (Rows, Columns)  
Number of rows: 72  
Number of columns: 18
```

- Number of Rows: 72 rows.
- Number of Columns: 18 columns.
- Time Range: From January 2020 to September 2025.

Population Data

```
--- Data Volume ---  
Shape of population_df: (71, 55) (Rows, Columns)  
Number of rows: 71  
Number of columns: 55
```

- Number of Rows: 71 rows (after initial cleaning).
- Number of Columns: 55 columns (after reorganization).
- Time Range: From January 2020 to November 2025.

BUT WE GOT THIS!

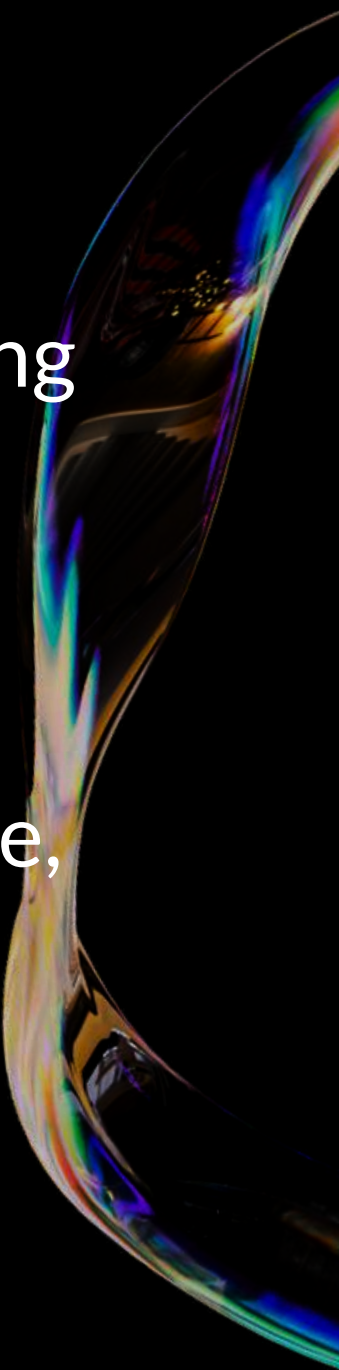
DATASETS WE PICKED

We already had the population and retail sales data for each province

HOW DID WE EXPAND? – BEFORE FEATURE ENGINEERING (PREPROCESSING)

Added features (columns) from other datasets we thought were related to retail spending (province-wise data only).

- Age-group based population - Divided into buckets of children (below 15), 15-29 (young adults), 30-59 (adults), 60+(senior citizens/retired)
- Age-based Unemployment Rate - Again, divided into the same age buckets as above, but does not include below 15 (because that will be child labor :o)
- No. of households in every province
- CPI - Consumer price index (to account for inflation)



HOW DOES THE DATASET LOOK NOW?

Huge increase in the data size, data variety, and features--

```
df.head()
```

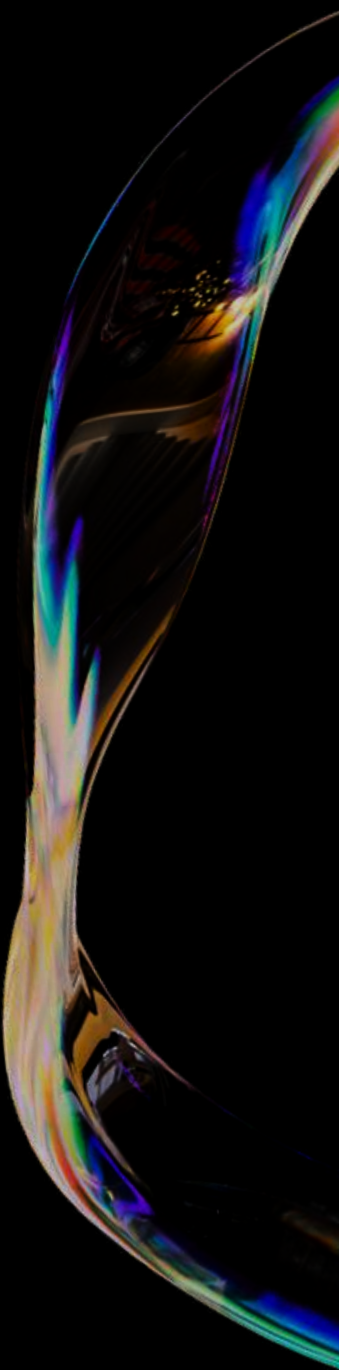
	Period	City_Province	Retail_Sales	Population	Population_Male	Population_Female	Population_15_29	Population_30_59	Population_60_Plus	Population_below_15	Unemployment_Rate_15_29	Unemployment_Rate_30_59	Unemployment_Rate_60_Plus	Number_
0	2020.01	Busan	504010	3411819.0	1674202.0	1737617.0	589397	1533328	918451	370643	9.2	2.4	6.1	
1	2020.01	Chungcheongbuk-do	104740	1599391.0	810129.0	789262.0	282119	717198	399784	200290	7.7	2.3	5.1	
2	2020.01	Chungcheongnam-do	153993	2122455.0	1081552.0	1040903.0	353199	947941	541133	280182	8.4	2.7	4.4	
3	2020.01	Daegu	360931	2436488.0	1204387.0	1232101.0	452592	1126167	563729	294000	9.7	2.5	4.9	
4	2020.01	Daejeon	195872	1474152.0	736259.0	737893.0	295260	685926	301906	191060	8.0	3.7	7.4	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
final_df.shape
```

(1207, 25)

It further increases once we do feature engineering (Covered under ML)



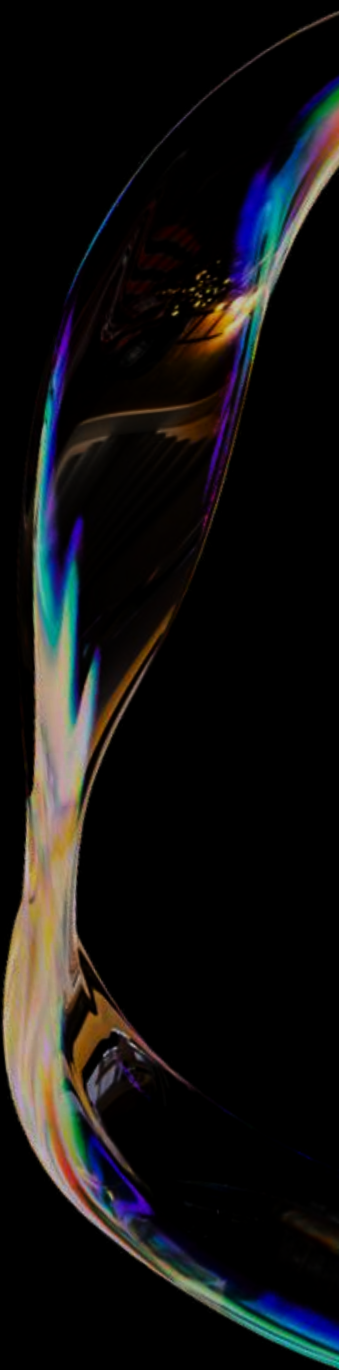
MISSING VALUES

df.isna().sum()

...	0
Period	0
City_Province	0
Retail_Sales	0
Population	0
Population_Male	0
Population_Female	0
Population_15_29	0
Population_30_59	0
Population_60_Plus	0
Population_below_15	0
Unemployment_Rate_15_29	0
Unemployment_Rate_30_59	0
Unemployment_Rate_60_Plus	0
Number_of_Households	0
Household_Size	0
Retail_Sales_Per_Household	0

dtype: int64

It appears that the Python script we used was highly effective in merging the data and creating the combined dataset, with no loss of data or missing values. Therefore, that is one issue averted



DATA EXPLORATION – IMPORTANT

Population Data

--- Population Change Analysis (Total, Male, Female) ---

Period: 2020-01 to 2025-11

Population Change by Region:

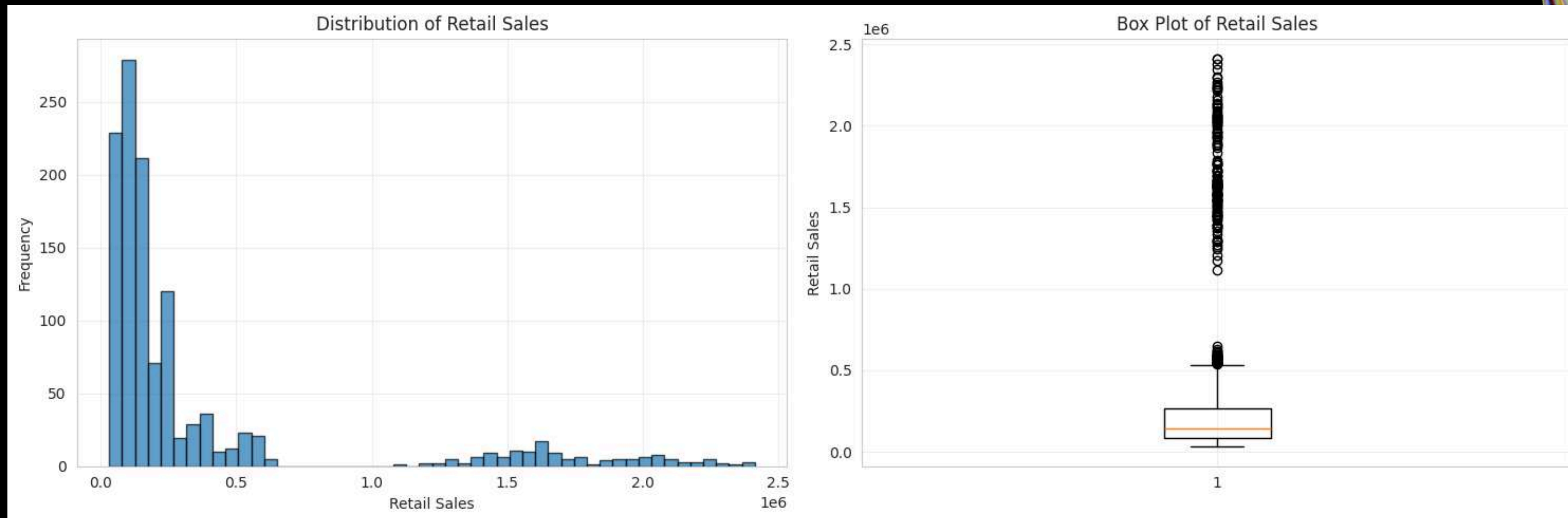
Region	Total Change	Total %	Male Change	Male %	Female Change	Female %
Busan	-168,060	-4.93%	-97,437	-5.82%	-70,623	-4.06%
Chungcheongbuk-do	-6,240	-0.39%	1,848	0.23%	-8,088	-1.02%
Chungcheongnam-do	14,511	0.68%	15,446	1.43%	-935	-0.09%
Daegu	-82,090	-3.37%	-50,995	-4.23%	-31,095	-2.52%
Daejeon	-32,266	-2.19%	-17,747	-2.41%	-14,519	-1.97%
Gangwon-do	-31,589	-2.05%	-16,025	-2.07%	-15,564	-2.03%
Gwangju	-62,387	-4.28%	-32,935	-4.57%	-29,452	-4.00%
Gyeonggi-do	475,342	3.59%	227,676	3.42%	247,666	3.76%
Gyeongsangbuk-do	-153,377	-5.76%	-70,501	-5.26%	-82,876	-6.27%
Gyeongsangnam-do	-152,484	-4.54%	-73,481	-4.34%	-79,003	-4.73%
Incheon	94,428	3.19%	42,020	2.84%	52,408	3.55%
Jeju-do	-5,827	-0.87%	-5,086	-1.51%	-741	-0.22%
Jeollabuk-do	-91,342	-5.03%	-43,481	-4.81%	-47,861	-5.24%
Jeollanam-do	-85,470	-4.58%	-37,271	-3.98%	-48,199	-5.19%
Sejong_si	50,167	14.65%	24,635	14.43%	25,532	14.88%
Seoul	-427,831	-4.40%	-260,484	-5.49%	-167,347	-3.35%
Ulsan	-54,464	-4.75%	-26,029	-4.42%	-28,435	-5.10%
Whole country	-718,979	-1.39%	-419,847	-1.62%	-299,132	-1.15%

--- Whole Country Summary ---

Category	Starting Pop	Ending Pop	Change	Percent
Total Population	51,847,509	51,128,530	-718,979	-1.39%
Male Population	25,862,863	25,443,016	-419,847	-1.62%
Female Population	25,984,646	25,685,514	-299,132	-1.15%

EXPLORATORY DATA ANALYSIS

Distribution of Retail Sales

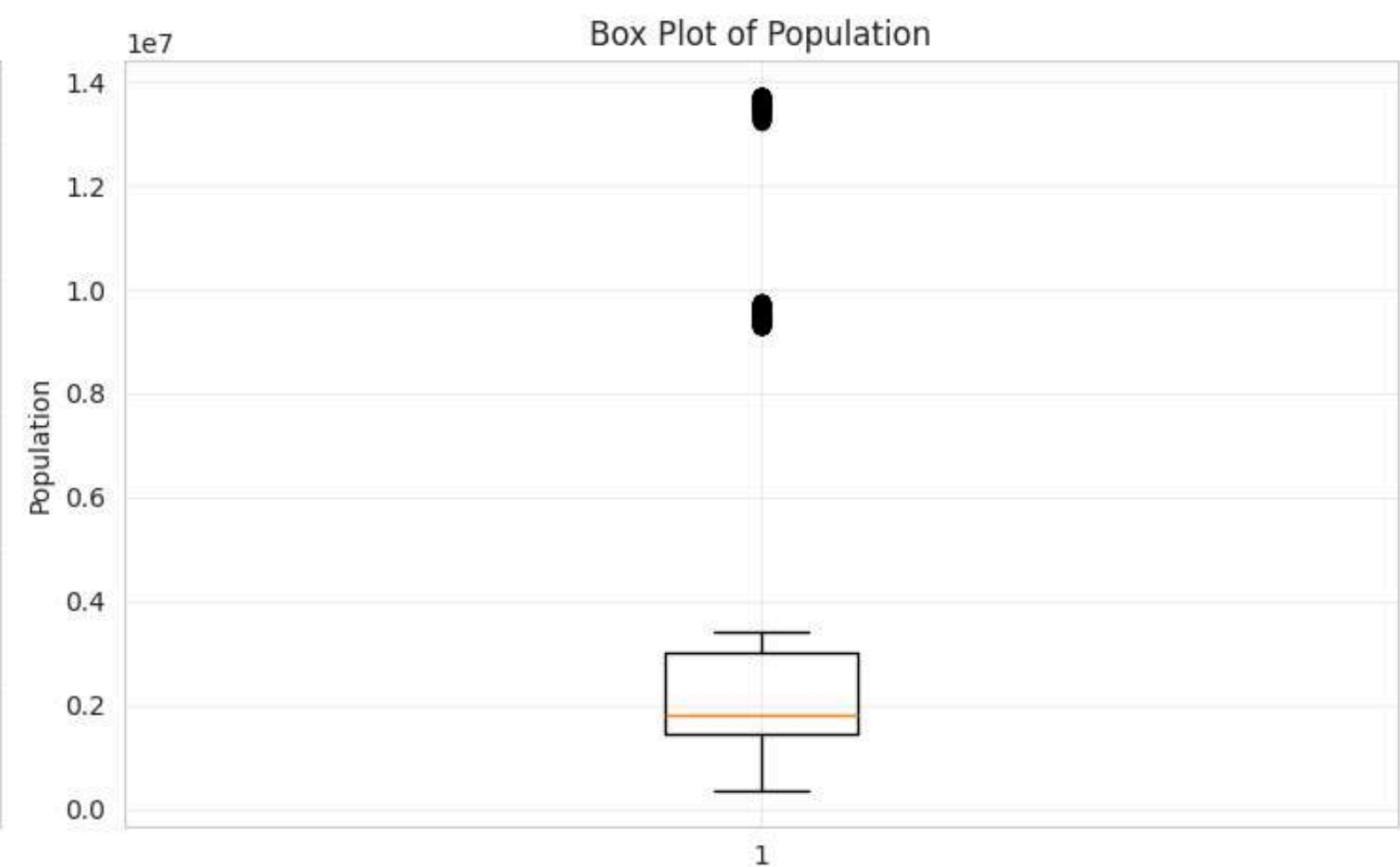
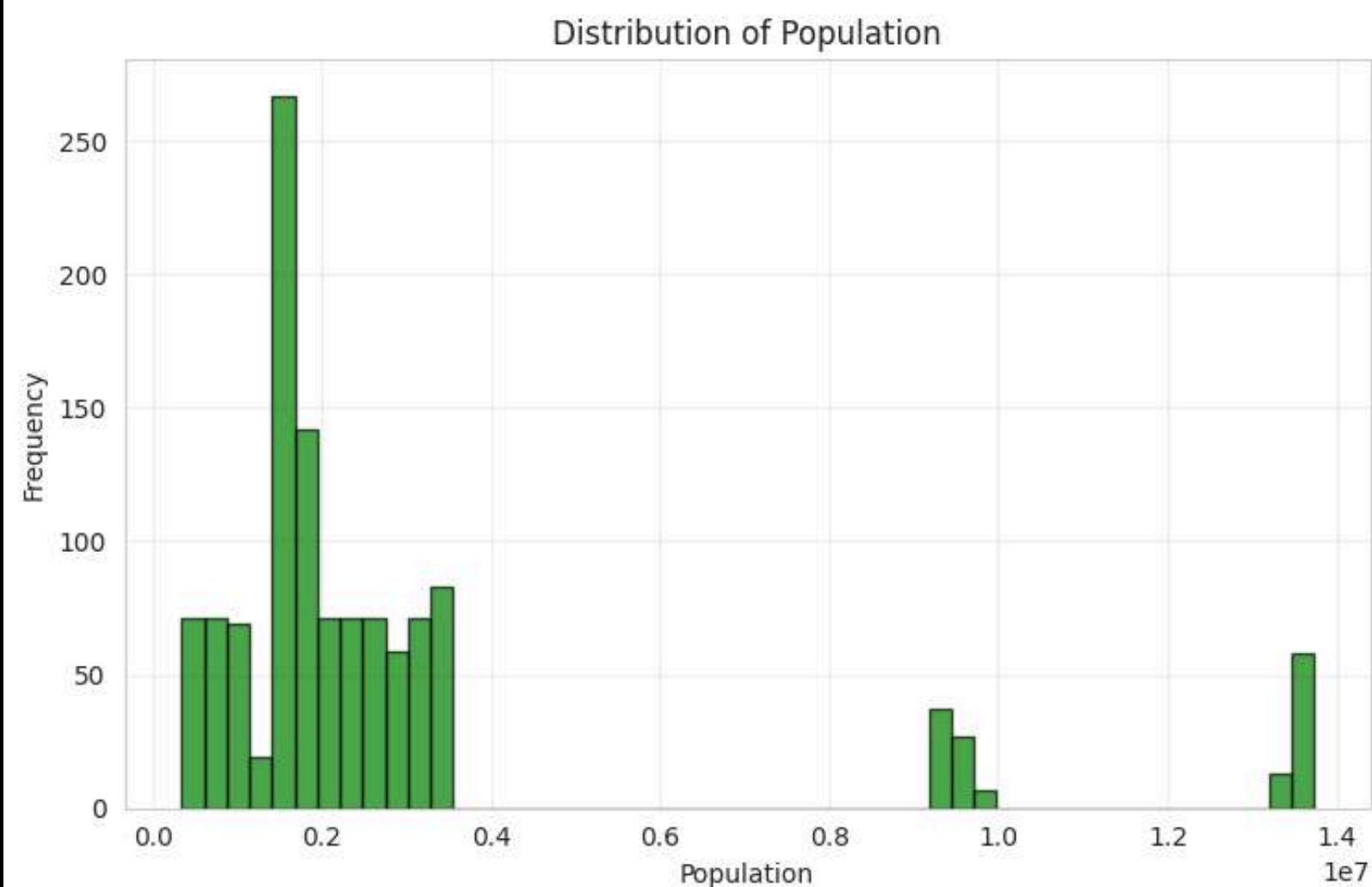
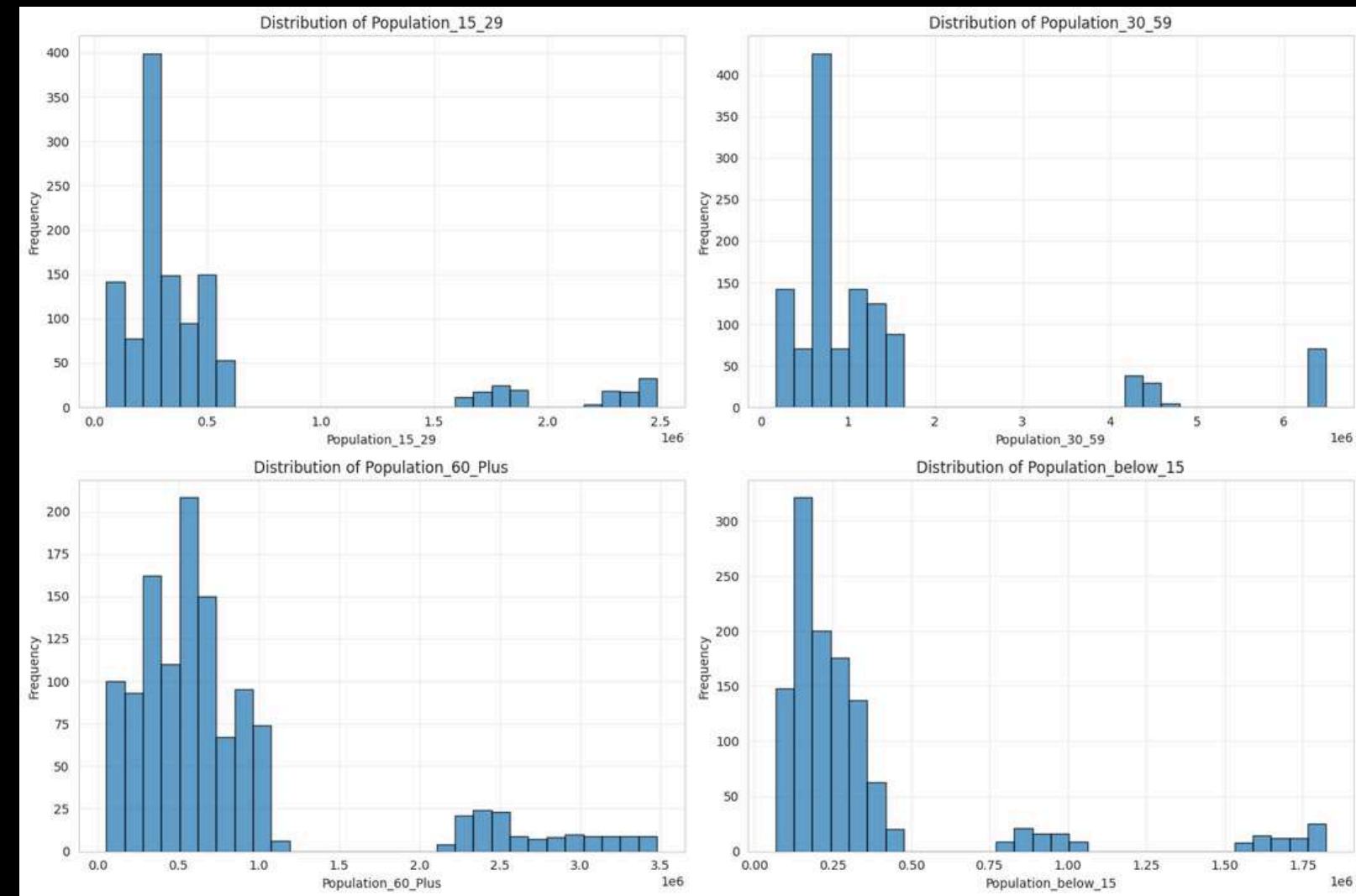


- Histogram: This plot shows a highly right-skewed distribution for Retail Sales. This means that a large number of cities/provinces have relatively low retail sales, while a few have extremely high sales, indicating significant disparity.
- Box Plot: The box plot confirms the right-skewness and highlights the presence of numerous outliers on the higher end, representing cities/provinces with exceptionally high retail activity.
- Summary: The mean (354,330.47) is much higher than the median (144,576.00), which is typical for right-skewed data and reinforces that most regions have lower sales, while a few major ones pull the average up.

EXPLORATORY DATA ANALYSIS

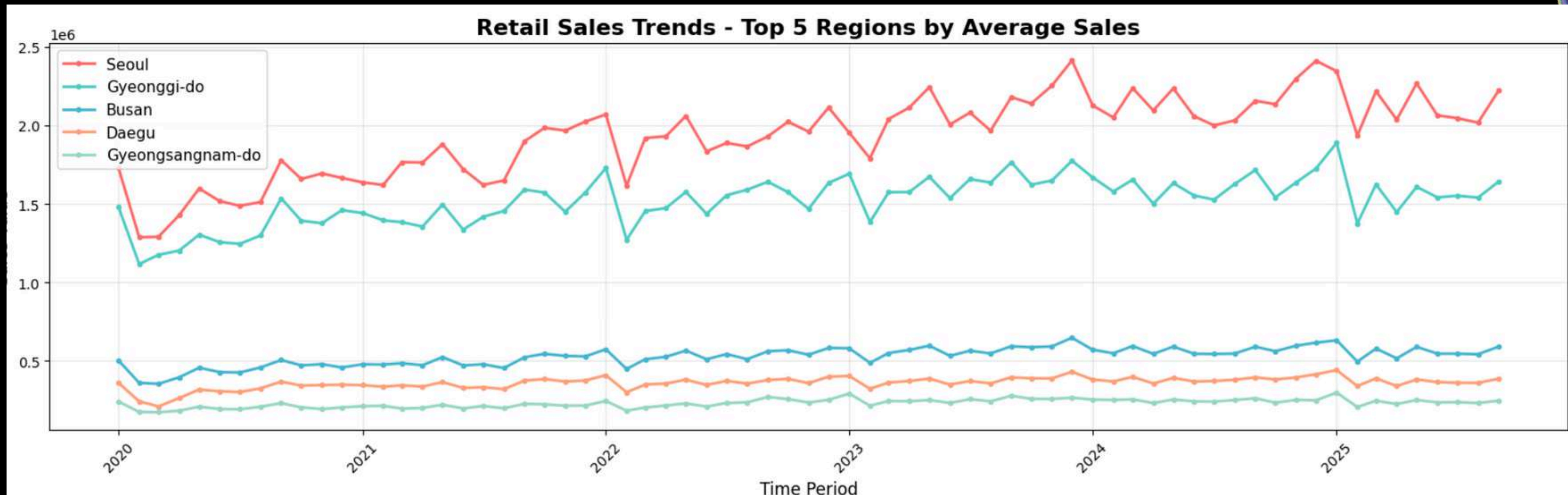
Distribution of Population

- Histogram & Box Plot: Similar to retail sales, the population distribution is also highly right-skewed. This indicates that most regions have smaller populations, while a few areas are significantly more populated, which is a common demographic pattern in many countries. The box plot also shows many outliers on the higher end.



EXPLORATORY DATA ANALYSIS

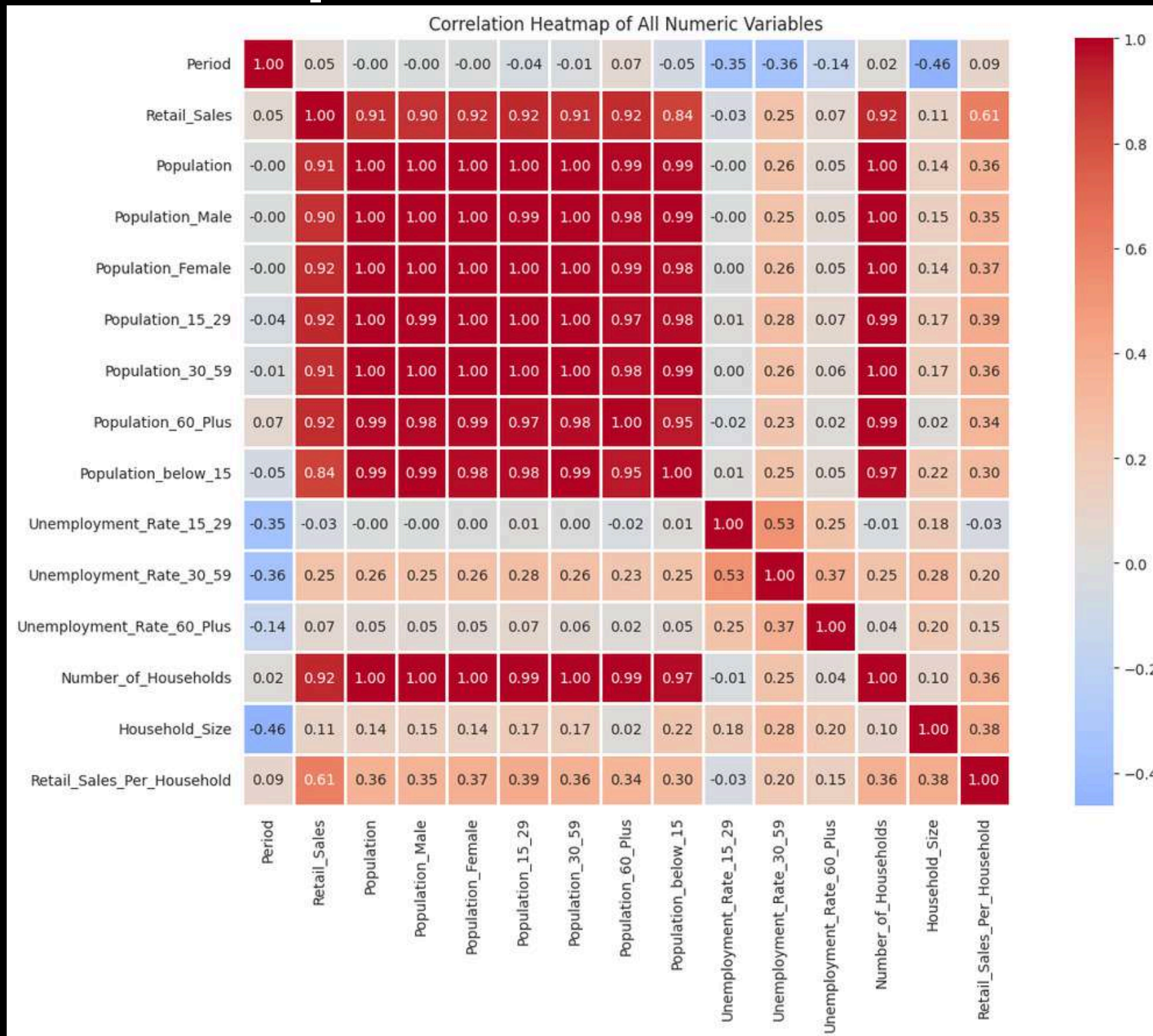
Distribution of Retail Sales



1. This chart clearly illustrates the significant disparity in average retail sales across different cities/provinces.
2. Regions like Seoul and Gyeonggi-do stand out with exceptionally high average retail sales, far surpassing other areas. This highlights the presence of major economic hubs that drive a large portion of the overall retail market.

EXPLORATORY DATA ANALYSIS

Heatmap

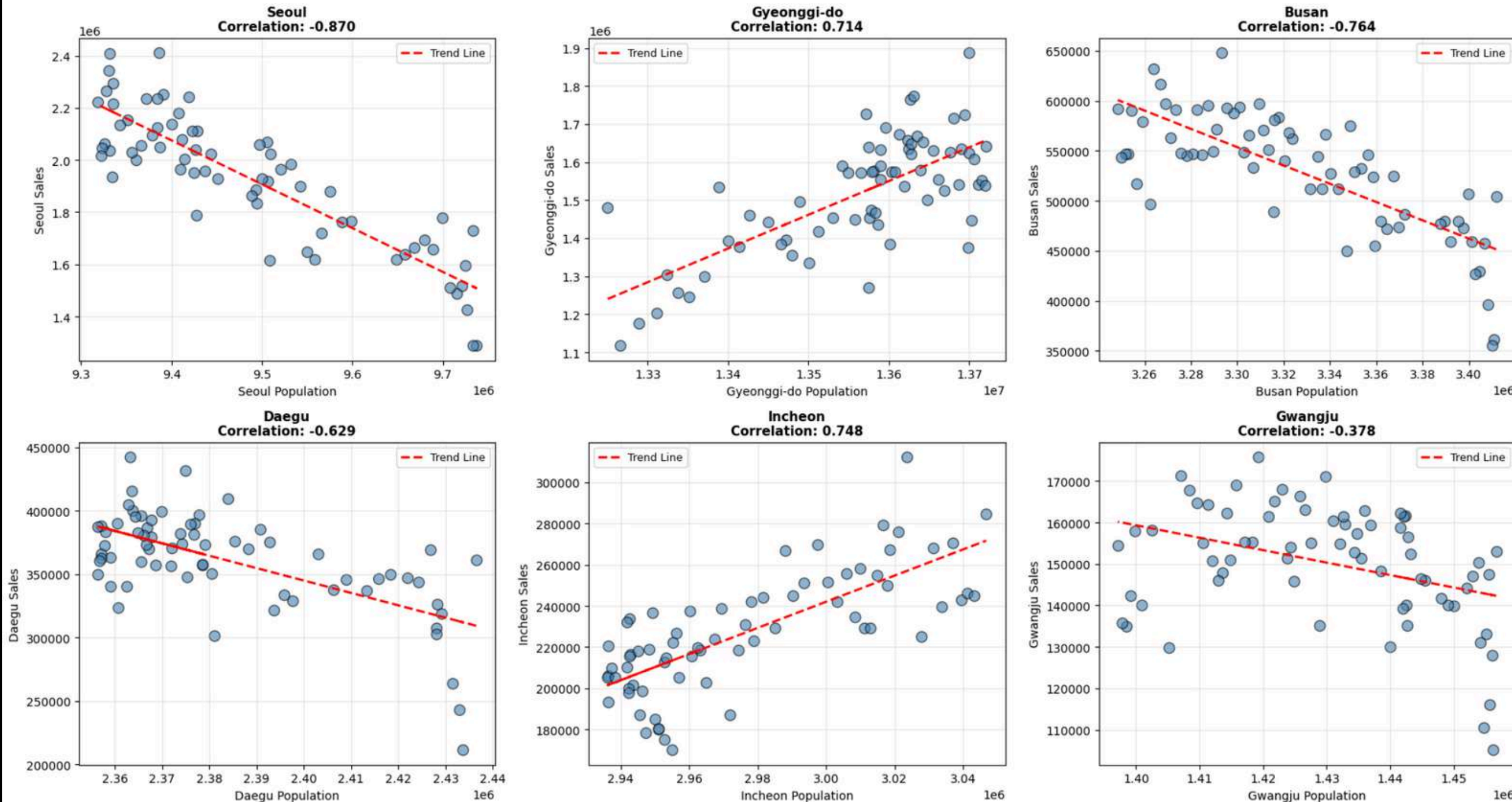


- Heatmap: This heatmap visually represents the correlation matrix between all numeric variables. It quickly allows us to identify strong positive (red hues) and negative (blue hues) relationships. We can observe strong positive correlations among most population-related features, number of households, and retail sales, reinforcing earlier findings. For instance, Retail_Sales is highly correlated with Population, Population_Male, Population_Female, Number_of_Households, and various age group populations.

EXPLORATORY DATA ANALYSIS

Population vs Retail Sales & Number of Households vs Retail Sales

Population vs Retail Sales Correlation by Region

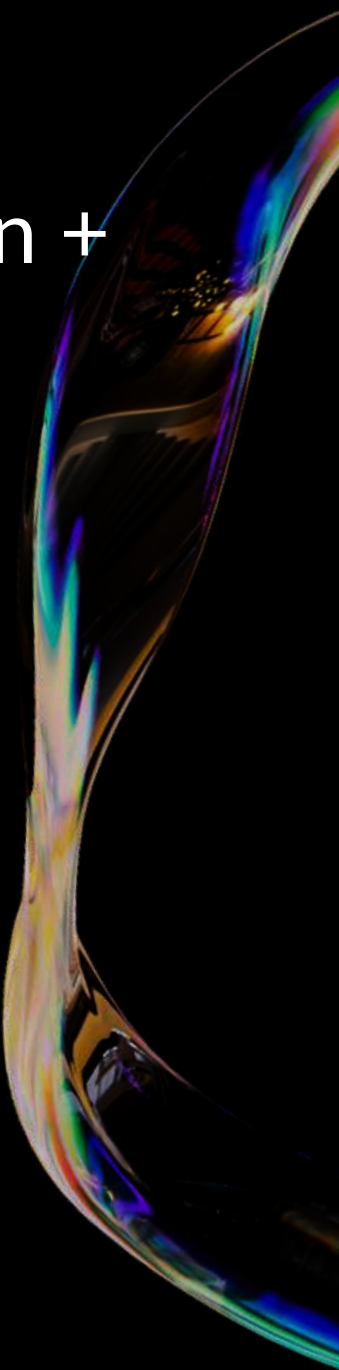


Inferences

1. Strong Positive Correlation- Incheon and Gyeonggi-do (as population increases, spending increases)
2. Strong Negative Correlation - Seoul, Busan, and Daegu (as population decreases, sales increases)
3. Weak Negative Correlation - Gwangju

MACHINE LEARNING FEATURE ENGINEERING

- Lag features: A column showing previous month sales to calculate growth
- Sales Month-on-month growth: Monthly percentage change in sales
- Sales Year-on-year growth: Yearly percentage change in sales
- Dependency Ratio: The “burden” on the productive part of the population ((Children + retired) / Working age population)
- The size of households (calculated based on total population/no. of households)
- Retail spending per household (calculated based on total retail spending/no. of households)
- Sales per capita (calculated based on total retail spending/ total population)

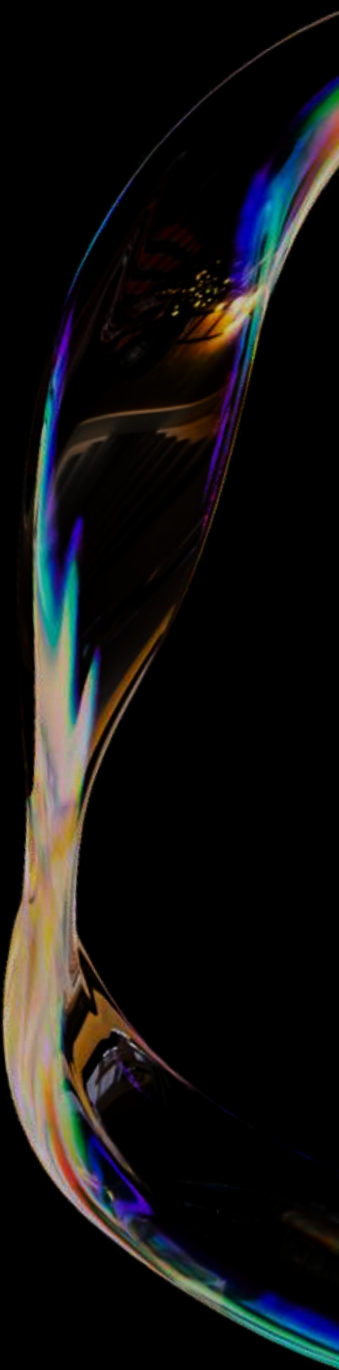


MISSING VALUES

Period	0
City_Province	0
Retail_Sales	0
Population	0
Population_Male	0
Population_Female	0
Population_15_29	0
Population_30_59	0
Population_60_Plus	0
Population_below_15	0
Unemployment_Rate_15_29	0
Unemployment_Rate_30_59	0
Unemployment_Rate_60_Plus	0
Number_of_Households	0
Household_Size	0
Retail_Sales_Per_Household	0
Date	0
Sales_Lag_1M	17
Sales_MoM_Growth	17
Sales_YoY_Growth	204
Working_Age_Pop	0
Dependency_Ratio	0
Sales_Per_Capita	0
Quarter	0
CPI	0

While there are no missing values from the raw datasets we inputted, due to growth rates being dependent on former data, there are no references in the first months or year for each district.

However, as our target variable output is the sales amount, these variables will not be included as part of our input.



MODEL COMPARISONS

--- Model Performance Comparison ---

	MAE	MSE	RMSE	R2 Score
Linear Regression	51607.506060	4.501748e+09	67095.070254	0.977014
Decision Tree	19324.495868	2.049202e+09	45268.116037	0.989537
Gradient Boosting	16397.836948	1.279968e+09	35776.639240	0.993464
Random Forest	18613.813926	1.901986e+09	43611.760432	0.990288

Inferences

1. Gradient Boosting has the smallest RMSE, meaning its predictions are closest to the actual values and higher predictive accuracy.
2. The numbers are extremely high, but the average total sales amount is 6 TRILLION, making the RMSE values fairly acceptable in this case
3. Gradient Boosting also has the highest R2 score, making it the best fit to our data

```
... — Gradient Boosting Regressor TRAIN Performance —  
RMSE (Train): 31416.911634116193  
R-squared (Train): 0.9967052424366117
```

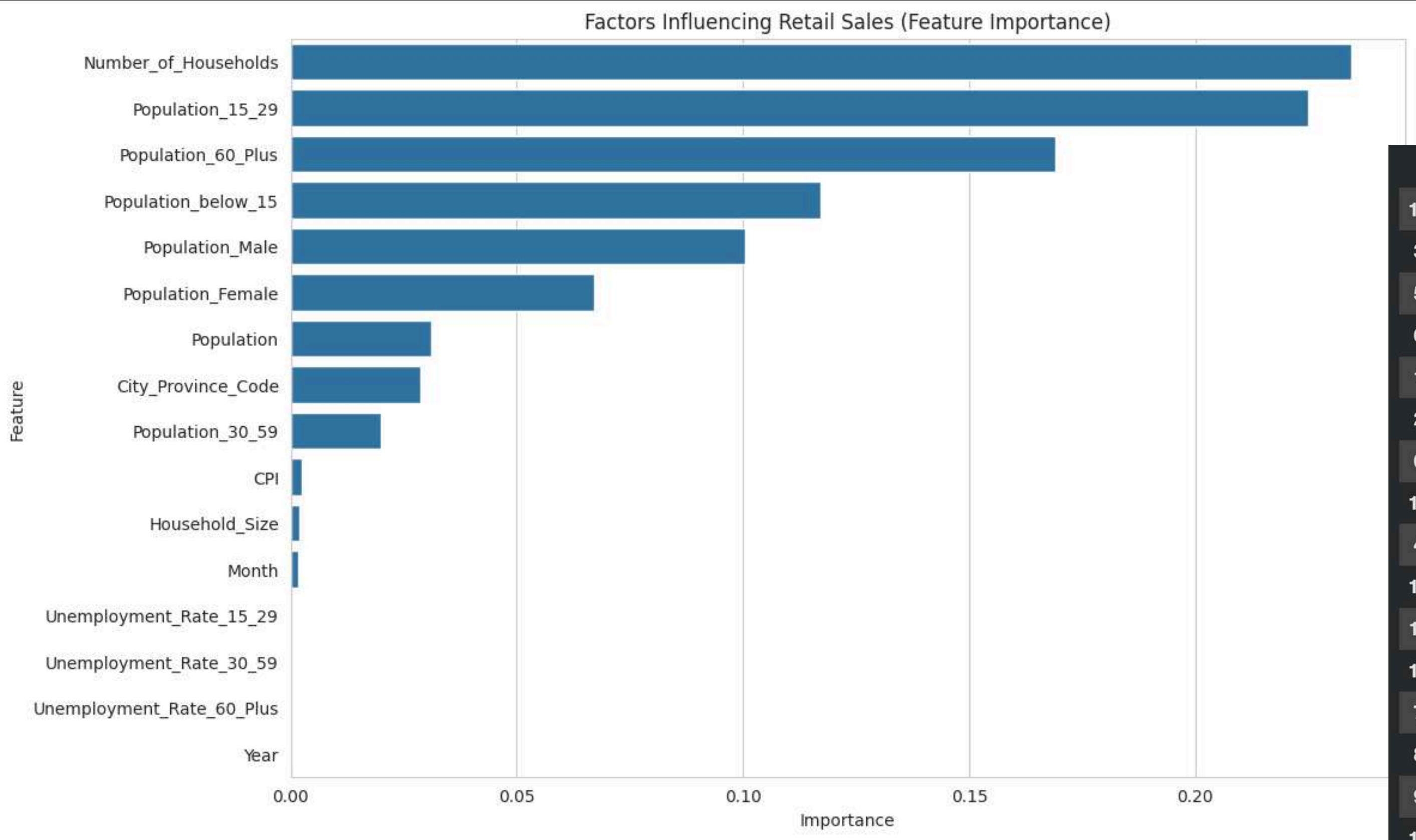

VISUALIZATION OF MODEL PERFORMANCE



Inferences

The predicted values for the test data mostly match or are in proximity to the actual historical data, indicating that the machine learning model is quite accurate

MOST IMPORTANT FACTORS



	Feature	Importance
10	Number_of_Households	0.234581
3	Population_15_29	0.225080
5	Population_60_Plus	0.169170
6	Population_below_15	0.117195
1	Population_Male	0.100433
2	Population_Female	0.067082
0	Population	0.031088
15	City_Province_Code	0.028638
4	Population_30_59	0.020011
12	CPI	0.002480
11	Household_Size	0.002006
14	Month	0.001758
7	Unemployment_Rate_15_29	0.000218
8	Unemployment_Rate_30_59	0.000143
9	Unemployment_Rate_60_Plus	0.000108
13	Year	0.000012

BUT WAIT.. I JUST SEE POPULATION EVERYWHERE

At first, we did too and thought something is wrong. However, upon diving deep, we came across a study performed by students from Incheon National University & Kyungpook National University

Demographic influences on urban commercial spaces: Spatial characteristics and purchase patterns across age groups in Incheon, Korea

Highlights

- Finds that purchasing patterns and environmental characteristics vary by age group.

Link -
<https://www.sciencedirect.com/science/article/pii/S026427512500143X?>



BUT WAIT.. I JUST SEE POPULATION EVERYWHERE

At first, we did too and thought something is wrong. However, upon diving deep, we came across a study performed by students from Incheon National University & Kyungpook National University

Demographic influences on urban commercial spaces: Spatial characteristics and purchase patterns across age groups in Incheon, Korea

Highlights

- Finds that purchasing patterns and environmental characteristics vary by age group.

Link -
<https://www.sciencedirect.com/science/article/pii/S026427512500143X?>



DISCUSSION OF LIMITATIONS

No unstructured data sources:

Our pipeline relies only on structured tables (retail sales, demographics, CPI, unemployment, households).

These were directly sourced from KOSIS, which makes it heavily reliant on to further improve the model in the future.

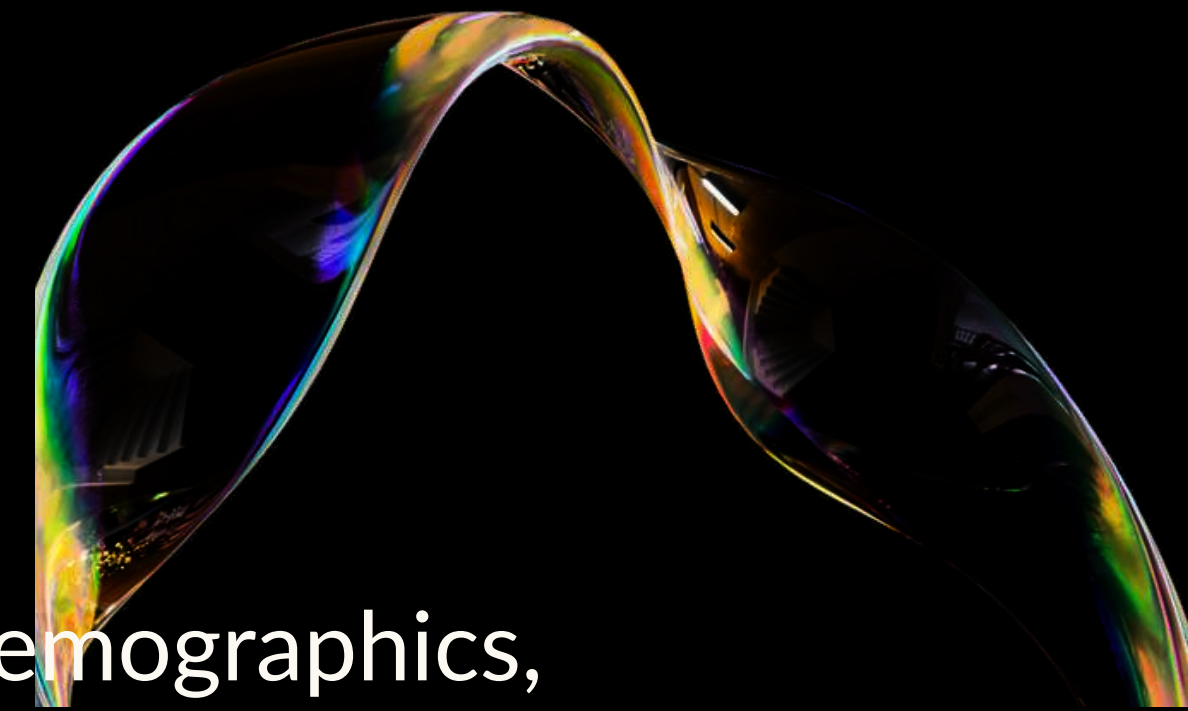
It does not capture qualitative or fast-moving factors such as:

**GOVERNMENT POLICY
CHANGES**

**CONSUMER
CONFIDENCE LEVEL**

SHOCKS (PANDEMICS)

**TAX/SUBSIDY
PROGRAMS**



CONCLUSION

- We built an end-to-end pipeline combining retail sales with demographic and macroeconomic indicators, trained a gradient boost model, and generated multi-year forecasts.
- The model captures relationships between retail sales and key drivers (especially household and population-related features) within the available dataset.
- However, the reliability of very long-term forecasts is constrained by limited history and missing external drivers.



FUTURE WORK

Add unstructured signals

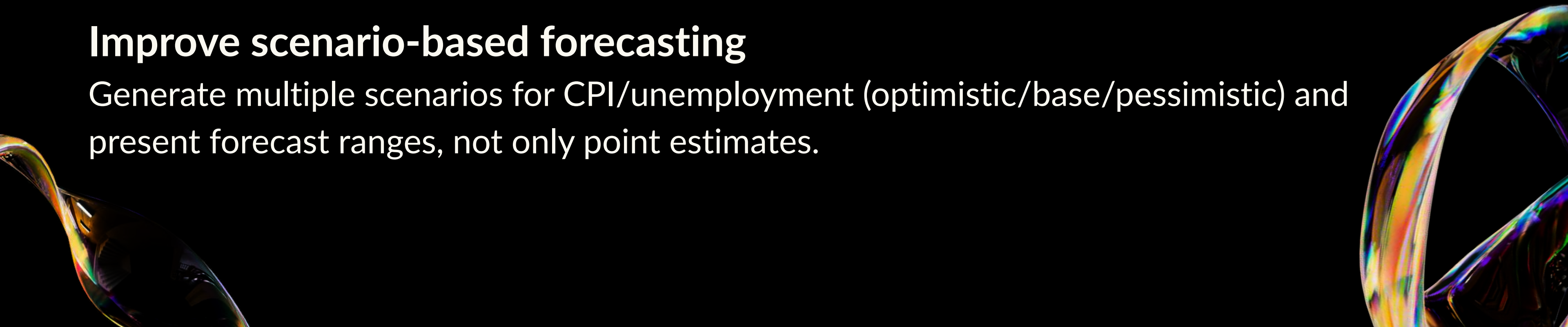
Incorporate policy/news indicators (economic policy timeline, sentiment scores) to better capture shocks and turning points.

Extend time coverage

Use a longer historical series to learn seasonality and business cycles more reliably.

Improve scenario-based forecasting

Generate multiple scenarios for CPI/unemployment (optimistic/base/pessimistic) and present forecast ranges, not only point estimates.





THANK YOU

for your time and attention

Presented by Team Data Blazers

